



Building a Spanish conversational system with router-controlled RAG

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Motivation & research questions

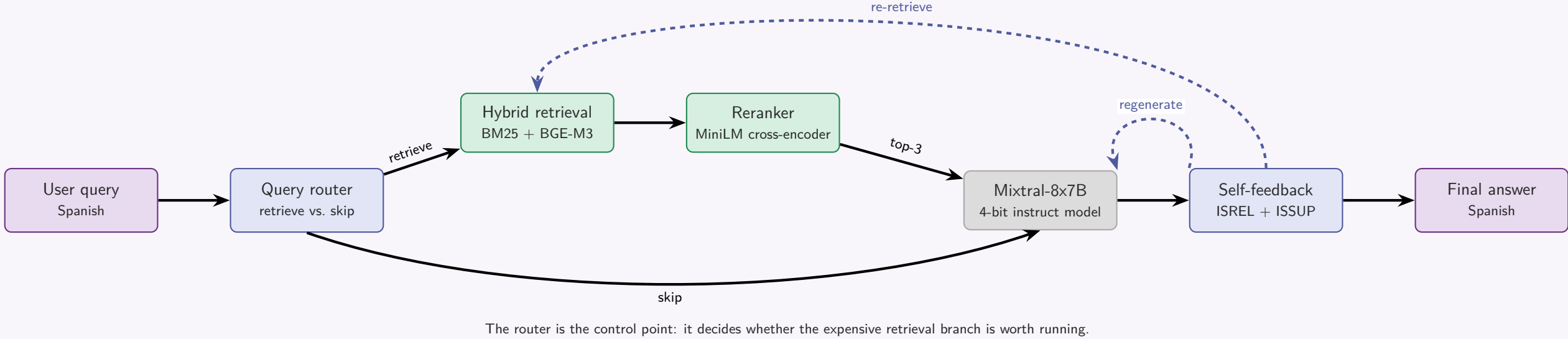
Retrieval-Augmented Generation (RAG) grounds factual answers in a corpus, but **always retrieving is wasteful**: chitchat (greetings, jokes, emotional support) neither needs nor benefits from corpus lookup.

This project builds a **Spanish conversational assistant** that first classifies each query with a router (**retrieve** or **skip**), then generates an answer with Mixtral-8x7B (Jiang et al., 2024), and finally undergoes an optional **self-feedback** step inspired by Self-RAG (Asai et al., 2023).

Research questions:

- RQ1: Can a learned router separate factual queries that need RAG from queries that do not?
- RQ2: Does routing preserve RAG answer quality while reducing latency?
- RQ3: Which router training strategy works best with fewest examples?
- RQ4: Does self-feedback repair incorrect RAG answers, unnecessary retrieval, or neither?

System pipeline



Datasets & data construction

Initial dataset with dataset labels

Source	Split	N samples	Label
SQAC	Router train	15,036	retrieve
ChatSubs	Router train	15,036	skip
SQAC	Router val	1,779	retrieve
ChatSubs	Router val	1,664	skip
SQAC	Test	85	retrieve
MS Chitchat	Test	85	skip

Generated oracle-labelled train sets

Source	Subset	Total	Per class
SQAC	Oracle-500	1,000	500
SQAC	Oracle-64	128	64

Oracle labels are produced by generating a RAG-based answer (*retrieve* class) and a RAG-skipped answer (*skip* class). The answer that obtains a higher BERT-F1 score against the gold answer is the oracle label of that sample.

Retrieval setup

- 5,601 SQAC passages
- BM25 + normalized BGE-M3 dense vectors.
- MiniLM reranker outputs top-3 passages
- **Critic prompt asks Mixtral for feedback**: `isrel` (whether retrieved context is relevant for the query) and `issup` (whether generated answer is supported by retrieved context).

Router training strategies

Router approaches

- **Frozen LR**: frozen multilingual MPNet embeddings + logistic regression. Trains in seconds. Strong baseline.
- **SetFit**: contrastive few-shot fine-tuning. Learns effectively from as few as 8 examples per class.
- **Fine-tune**: full encoder fine-tuning with HuggingFace Trainer (5 epochs, early stopping).

Metrics

- Router evaluation: oracle accuracy without self-feedback.
- Self-feedback (SF) evaluation: oracle accuracy without SF vs with SF (Δ).
- Answer quality evaluation: token overlap F1 (lexical quality) and multilingual BERTScore F1 (semantic quality).
- Cost evaluation: mean latency (temporal cost) and retrieval rate (computational cost).

Finding 1: Routing preserves quality at lower cost

Model	Retrieval rate ↓	Latency ↓	Token-F1 ↑	BERT-F1 ↑
Always RAG	100.0%	13.94s	7.88	61.10
Never RAG	0.0%	6.64s	4.57	58.21
Frozen LR router	48.2%	10.33s	7.39	60.81
SetFit router	48.8%	10.05s	8.12	61.22
Fine-tuned router	50.6%	9.98s	7.64	61.09
Oracle SetFit-500	47.6%	9.82s	6.98	59.78
Oracle SetFit-64	44.1%	10.10s	6.03	59.68

Answer quality. SetFit is the only router model that beats Always RAG. It performs slightly better in both token-F1 and BERT-F1.

Cost. All router models outperform the Always RAG baseline in retrieval rate and latency. SetFit reduces retrieval rate from 100% to 48.8% and mean latency from 13.94s to 10.05s (27.9% reduction).

Finding 3: Dataset accuracy does not correspond to oracle accuracy

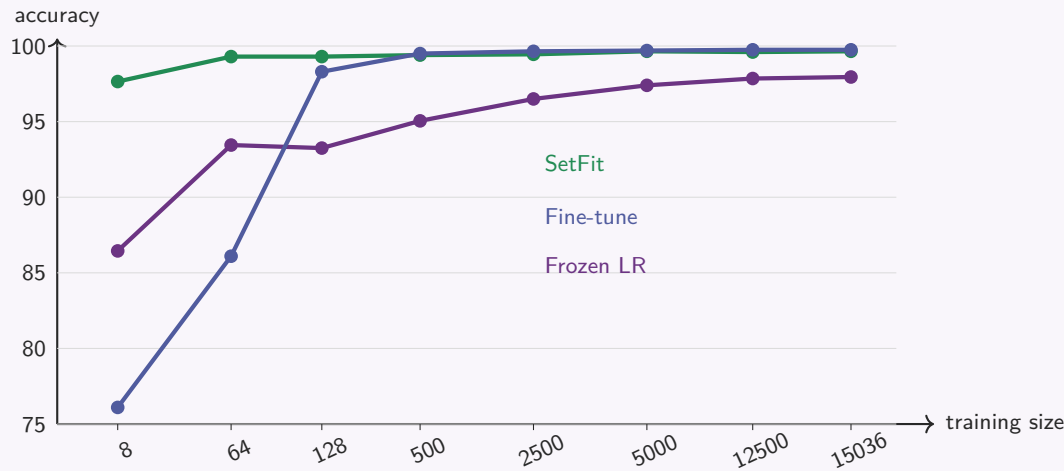
Model	Dataset accuracy ↑	Oracle accuracy ↑
Always RAG	50.00	69.41
Never RAG	50.00	30.59
Frozen LR router	97.06	67.06
SetFit router	98.82	69.41
Fine-tuned router	99.41	69.41
Oracle SetFit-500	52.94	57.65
Oracle SetFit-64	52.94	49.41

Routers trained on dataset labels: they achieve near-perfect performance in the dataset-label task. They do not perform so well in the real-world scenario (oracle-label task).

Routers trained on oracle labels (Oracle SetFit-64/500): they obtain worse results than routers trained on dataset labels, probably due to oracle data scarcity.

Subset	Oracle retrieval
SQAC	69.4%
Chitchat	0.0%
Overall	34.7%

Finding 2: The router learns the dataset labels quickly



Learning curve. SetFit reaches **99.31% validation accuracy with only 64 examples per class**. Full fine-tuning catches up after 500 examples per class and reaches the highest validation score at 12,500 examples per class.

Finding 4: Self-feedback (SF) improves oracle acc. and corrects over-retrieval

Model	Retrieval rate ↓			Latency ↓			Dataset acc. ↑			Oracle acc. ↑		
	NoSF	SF	Δ	NoSF	SF	Δ	NoSF	SF	Δ	NoSF	SF	Δ
Always RAG	100.0	61.8	-38.2	6.21s	13.94s	+7.72s	50.00	61.18	+11.18	69.41	77.65	+8.24
Frozen LR router	48.2	34.1	-14.1	5.82s	10.33s	+4.51s	97.06	84.12	-12.94	67.06	72.94	+5.88
SetFit router	48.8	35.9	-12.9	5.54s	10.05s	+4.51s	98.82	85.88	-12.94	69.41	76.47	+7.06
Fine-tuned router	50.6	37.6	-12.9	5.53s	9.98s	+4.45s	99.41	87.65	-11.76	69.41	75.29	+5.88
Oracle SetFit-500	47.6	32.4	-15.3	6.77s	9.82s	+3.05s	52.94	58.82	+5.88	57.65	57.65	+0.00
Oracle SetFit-64	44.1	22.9	-21.2	6.75s	10.10s	+3.35s	52.94	57.65	+4.71	49.41	49.41	+0.00

Correction of over-retrieval. In Always RAG, SF lowers retrieval rate from 100% to 61.8%. Interestingly, SF also lowers retrieval rate in router models: around 13% reduction.

Oracle accuracy improvement. Always RAG achieves the highest score but is closely followed by SetFit, which is much more efficient in retrieval rate and latency.

Qualitative analysis

Good skip: conversational query

Query: "Dime si puedes estornudar"

Route: skip retrieval

Answer: "No, no puedo estornudar porque soy un asistente digital..."

Effect: direct answer without forcing irrelevant retrieved context.

Good retrieve: factual QA

Query: "¿Qué cargos ocupa Jesús Gil?"

Route: retrieve SQAC passages

Answer: "alcalde de Marbella y presidente del Atlético de Madrid"

Effect: retrieved context supports a concise gold-like answer.

Good routing but unsuccessful retrieval

Query: "¿Qué es el SITE?"

Gold: "grupo de inteligencia"

Route: retrieve → fallback

Answer: "SITE se puede referir a a Society of Inventors and Technical Executives..."

Effect: retrieval failed to recover relevant SQAC evidence.

Main insight: the router reliably separates conversational and factual queries. The main bottleneck is retrieval quality, not the routing decision itself.

Take-aways

1. **A router is worthwhile:** router models obtain the same answer quality with a **considerable reduction in computational cost** (−51.2% retrieval rate) **and temporal cost** (−27.9% latency) relative to Always RAG.
2. **SetFit is the practical router choice.** It trains fast and effectively (**99% validation accuracy with only 64 examples per class**). Fine-tuning reaches same validation score at larger training sizes. Frozen LR lags behind.
3. **Oracle-label training is harder than dataset-label training.** Routers trained on oracle labels underperform the dataset-label routers in oracle accuracy, probably due to oracle data scarcity.
4. **Self-feedback boosts oracle accuracy**, especially in the Always-RAG model. Still, **SetFit** is the **best model** since it achieves **almost the same oracle accuracy as Always-RAG with lower retrieval rate and latency**.

Limitations & future work

Limitations

1. Oracle labels are model dependent: no easy solution, each LLM is pretrained on different data.
2. Fallback (i.e. when SF cancels retrieval) can produce ungrounded answers: solution to this limitation in Future work section.

Future work

1. Extend the oracle labels corpus size.
2. In order to tackle ungrounded caused by SF fallback: add **grounding-aware abstention**, i.e. if retrieval confidence or critic support is too low, refuse to answer. It is better to not answer than to hallucinate.

References:

Asai et al. (2023). *Self-RAG: Learning to Retrieve, Generate, and Critique through Self-Reflection*. arXiv:2310.11511
Jiang et al. (2024). *Mixtral of Experts*. arXiv:2401.04088